

```
def euler_method(f, x0, t0, dt, num_step):
```

```
    t = [t0]
```

```
    x = [x0]
```

```
    t_1 = t0
```

```
    x_1 = x0
```

```
    for _ in range(num_step):
```

```
        dx_dt = f(t_1, x_1)
```

```
        x_2 = x_1 + dt*(dx_dt)
```

```
        t_2 = t_1 + dt
```

```
        t.append(t_2)
```

```
        x.append(x_2)
```

```
        t_1 = t_2
```

```
        x_1 = x_2
```

```
    return t,x
```